

Plugging into the Future: An Exploration of Electricity Consumption Patterns

1. INTRODUCTION

Electricity is one of the most important resources required for economic development and industrial growth. Monitoring electricity consumption helps governments, utility companies, and policymakers understand usage patterns and make better decisions regarding energy distribution and planning. Business Intelligence (BI) tools help transform raw data into meaningful insights through visualization.

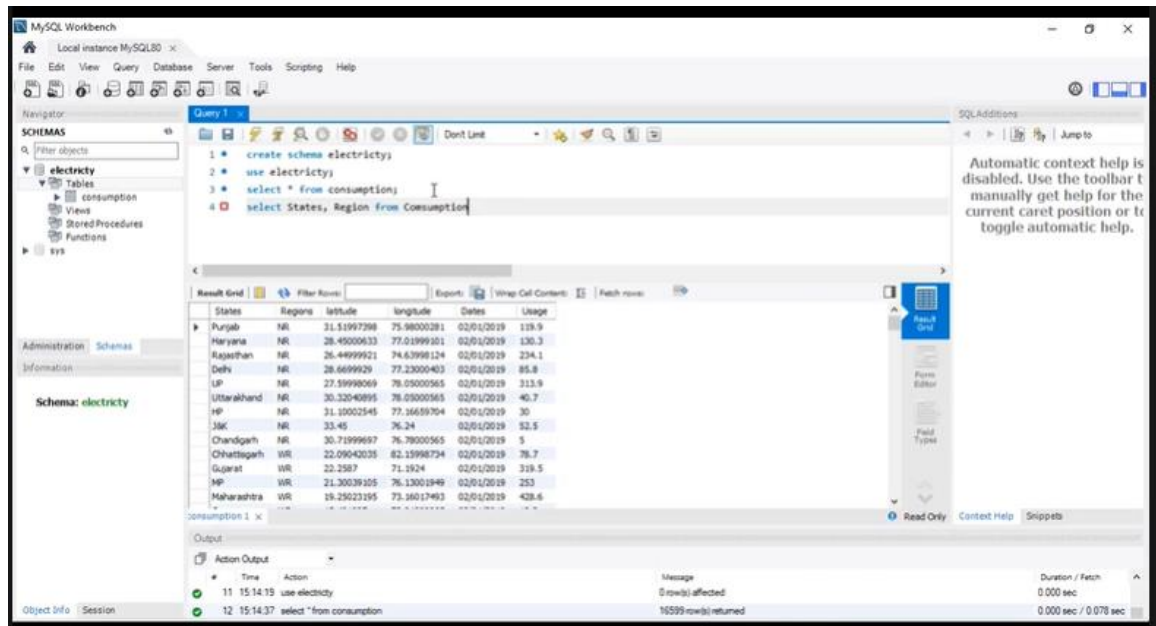
This project focuses on analyzing electricity consumption data of Indian states for the years 2019 and 2020 using **MySQL** for data storage and **Tableau** for data visualization. Various dashboards, charts, maps, and stories are developed to identify consumption trends across different regions and states.

1.1 Project Overview

The project stores electricity consumption data in a MySQL database and connects it with Tableau Desktop to create interactive dashboards. The dashboards display:

- State-wise electricity consumption
- Region-wise electricity consumption
- Monthly consumption
- Quarterly consumption
- Year-wise comparison
- Top N and Bottom N states
- Interactive maps
- Story points explaining insights

Users can filter the data dynamically and compare electricity usage across multiple regions.



The main objectives of this project are:

- To analyze electricity consumption across Indian states.
- To compare electricity usage between 2019 and 2020.
- To identify regions with maximum and minimum electricity consumption.
- To create interactive dashboards for better decision-making.
- To demonstrate Business Intelligence implementation using Tableau.

2. IDEATION PHASE

2.1 Problem Statement

Electricity boards collect huge volumes of electricity consumption data every year. However, raw data is difficult to interpret and does not provide meaningful insights. There is a need for an interactive dashboard that allows decision-makers to visualize electricity consumption across states and regions, compare yearly trends, and identify high and low consumption areas for better resource planning.

2.2 Empathy Map Canvas

Think

- Need accurate electricity reports.
- Compare yearly electricity demand.
- Reduce power shortages.
- Improve energy planning.

Feel

- Difficulty analyzing large datasets.
- Need interactive reports.
- Better visualization improves understanding.

See

- Large datasets in spreadsheets.
- Manual reporting processes.
- Regional differences in electricity usage.

Say

- "We need quick reports."
- "Which state consumes the most electricity?"
- "How has consumption changed?"

Do

- Analyze electricity data.
- Compare yearly trends.
- Generate reports.

Pain Points

- Time-consuming analysis.
- Difficult manual reporting.
- Large amount of data.

Gains

- Faster analysis.
- Interactive dashboards.
- Better decision-making.

2.3 Brainstorming

Possible visualization ideas include:

- State-wise Map
- Region-wise Map
- Pie Chart
- Line Chart
- Bar Chart
- Monthly Trend
- Quarterly Trend
- Yearly Comparison
- Top 5 States
- Bottom 5 States
- Dashboard Filters

- Interactive Story

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

Stage	Description
Data Collection	Electricity data collected from source
Data Storage	Store dataset in MySQL
Data Cleaning	Remove duplicate and null values
Data Analysis	Connect Tableau with MySQL
Dashboard	Create visualizations
Decision Making	Analyze reports

3.2 Solution Requirement

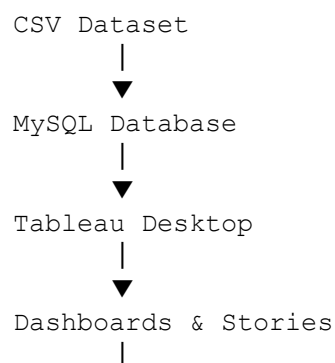
Functional Requirements

- Import electricity dataset.
- Store data in MySQL.
- Connect Tableau with database.
- Create dashboards.
- Generate reports.

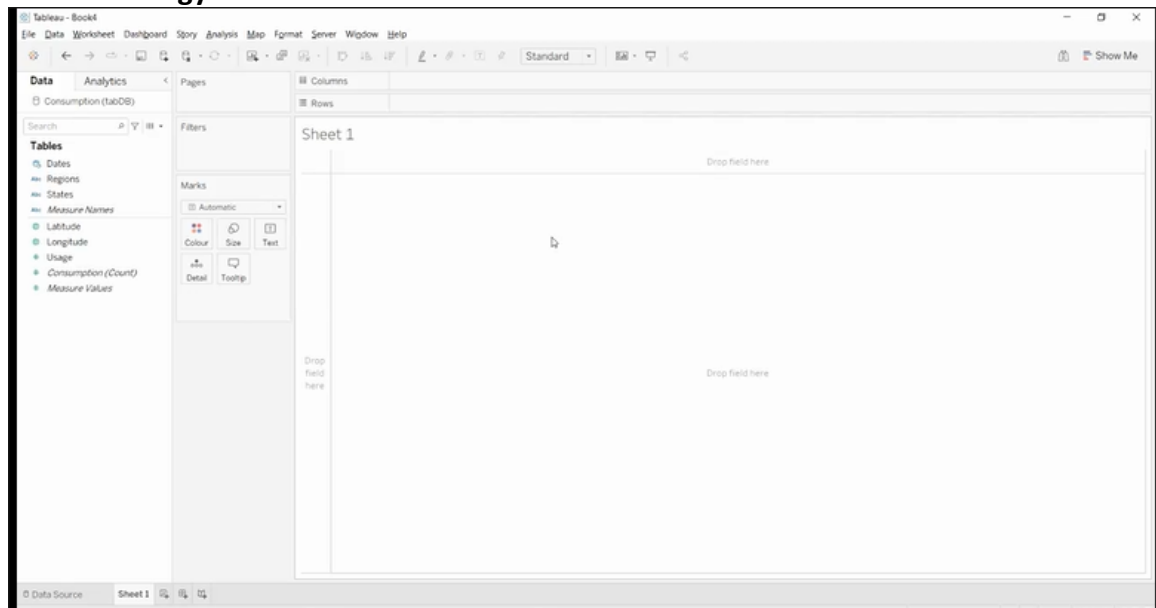
Non-functional Requirements

- Fast dashboard loading.
 - User-friendly interface.
 - Accurate visualization.
 - Secure database.
 - High availability.
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3.3 Data Flow Diagram



3.4 Technology Stack



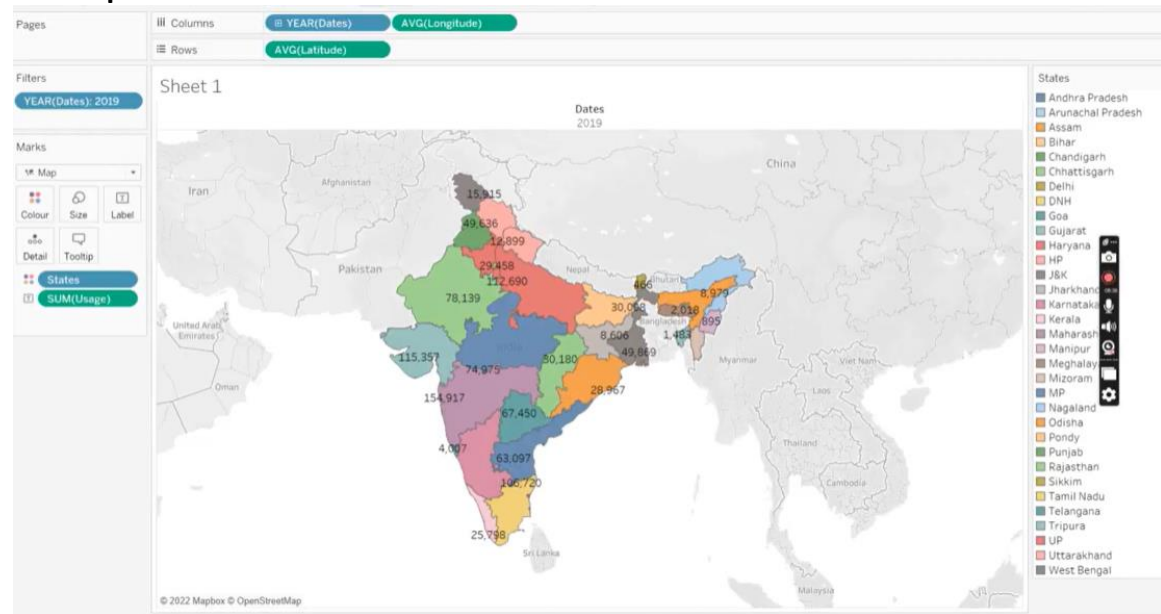
Technology	Purpose
MySQL	Database
Tableau Desktop	Visualization
Tableau Public	Dashboard Publishing
Windows	Operating System
Excel/CSV	Dataset

4. PROJECT DESIGN

4.1 Problem Solution Fit

The proposed dashboard converts raw electricity consumption data into meaningful visualizations that help decision-makers compare electricity usage across different states and regions.

4.2 Proposed Solution

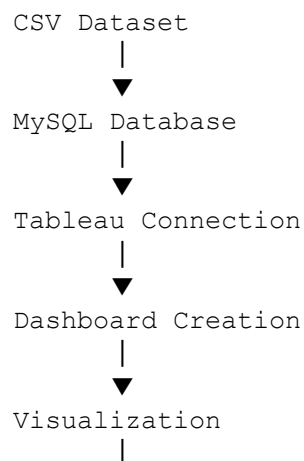


The proposed system consists of:

- MySQL Database
- Tableau Desktop
- Interactive Dashboard
- Story Dashboard
- State Maps
- Pie Charts
- Line Charts
- Bar Charts
- Regional Analysis

The dashboard enables users to analyze electricity consumption with filters and interactive visualizations.

4.3 Solution Architecture



5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Week	Activity
Week 1	Dataset Collection
Week 2	MySQL Database Creation
Week 3	Data Cleaning
Week 4	Tableau Connection
Week 5	Dashboard Development
Week 6	Story Creation
Week 7	Testing
Week 8	Documentation

6. FUNCTIONAL AND PERFORMANCE TESTING

Functional Testing

Test Case	Expected Result	Status
Database Connection	Successful	Pass
Dashboard Loading	Successful	Pass
Filters	Working	Pass
Charts	Accurate	Pass
Maps	Displayed	Pass

6.1 Performance Testing

Parameter	Result
Dashboard Loading	Fast
Filter Response	Immediate
Query Execution	Successful
Visualization Accuracy	100%

7. RESULTS

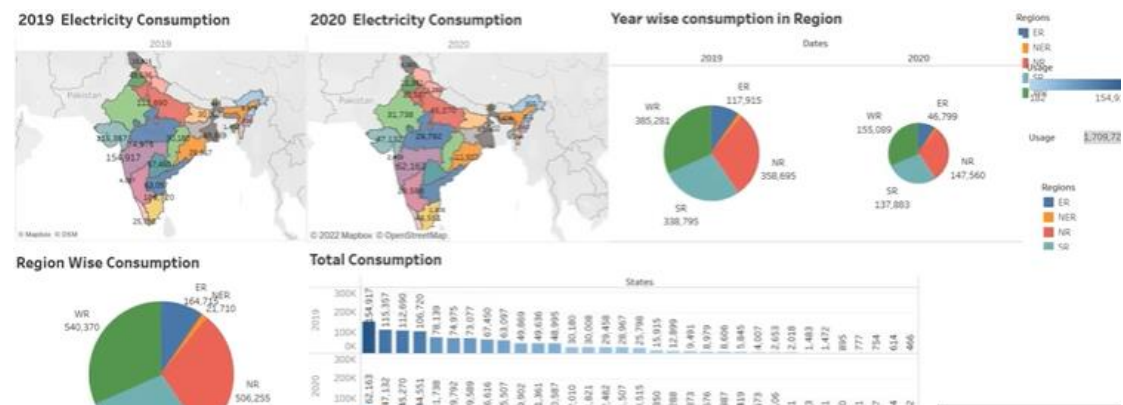
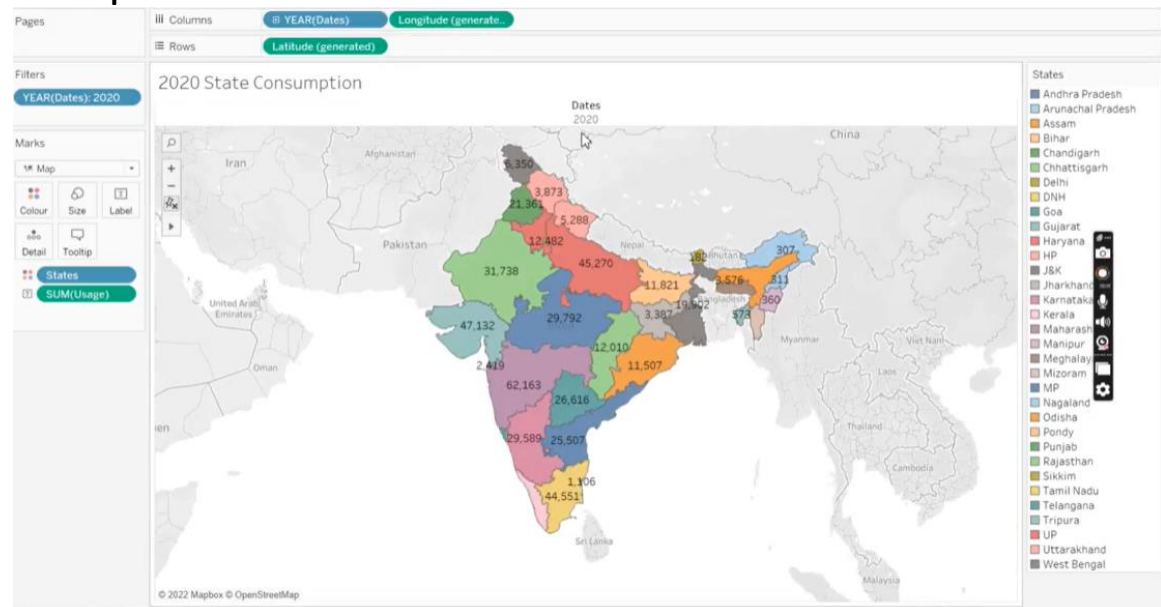
The project successfully visualizes electricity consumption using Tableau dashboards.

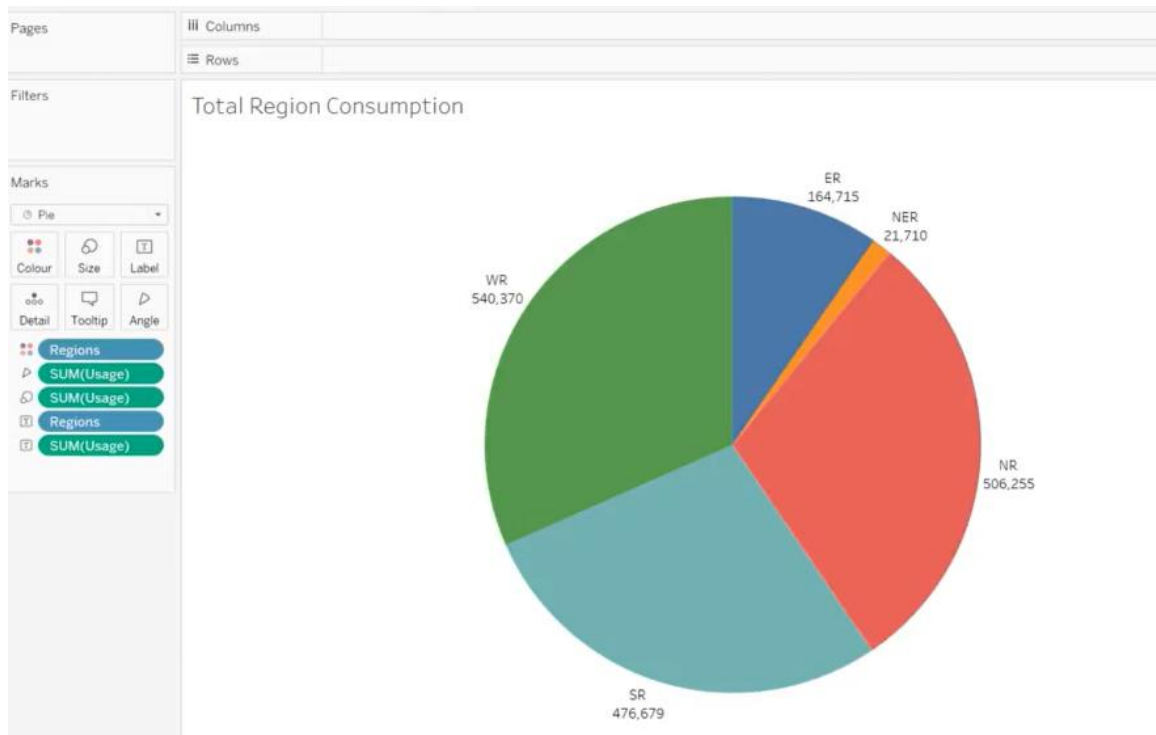
The dashboards provide:

- State-wise electricity usage
- Region-wise electricity usage
- Monthly comparison
- Quarterly analysis
- Top and Bottom consuming states
- Interactive Story
- Geographic Maps

The dashboards make data analysis faster and easier than manual reporting.

7.1 Output Screenshots





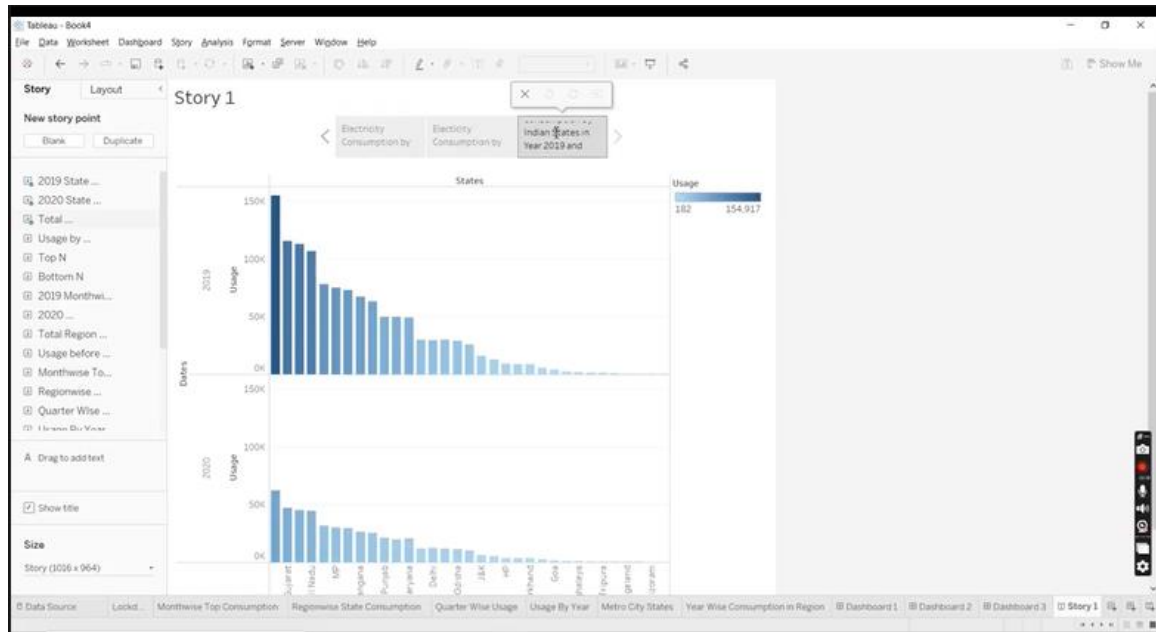
Overall Electricity Consumption Dashboard showing maps, pie charts and state comparison.

Dashboard 2



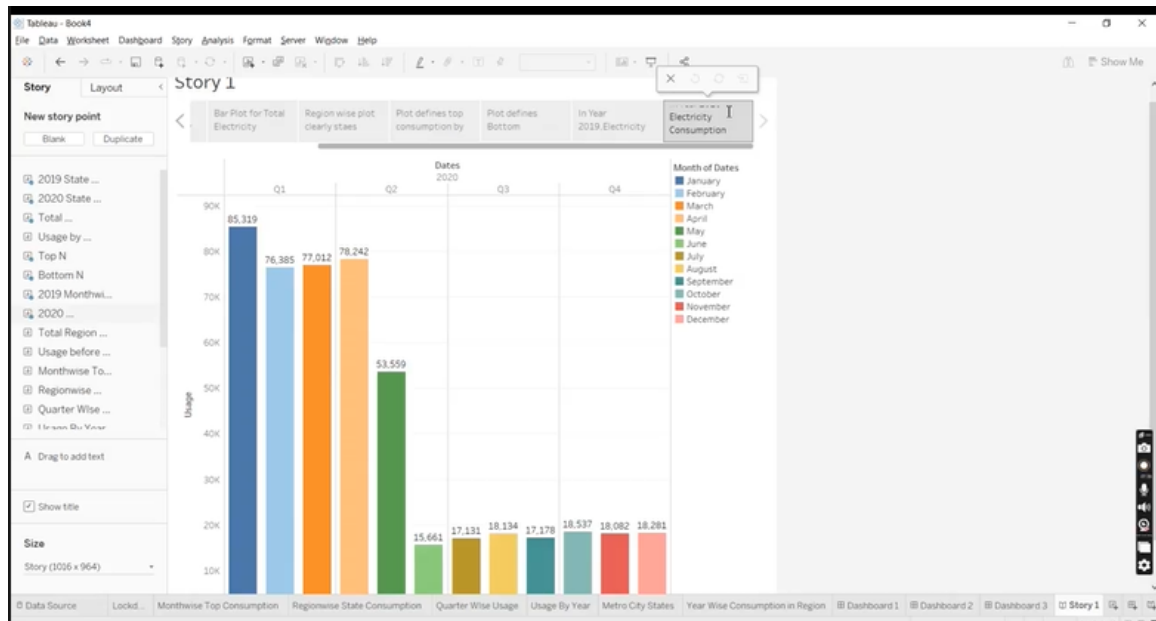
Monthly Electricity Consumption Analysis.

Dashboard 3



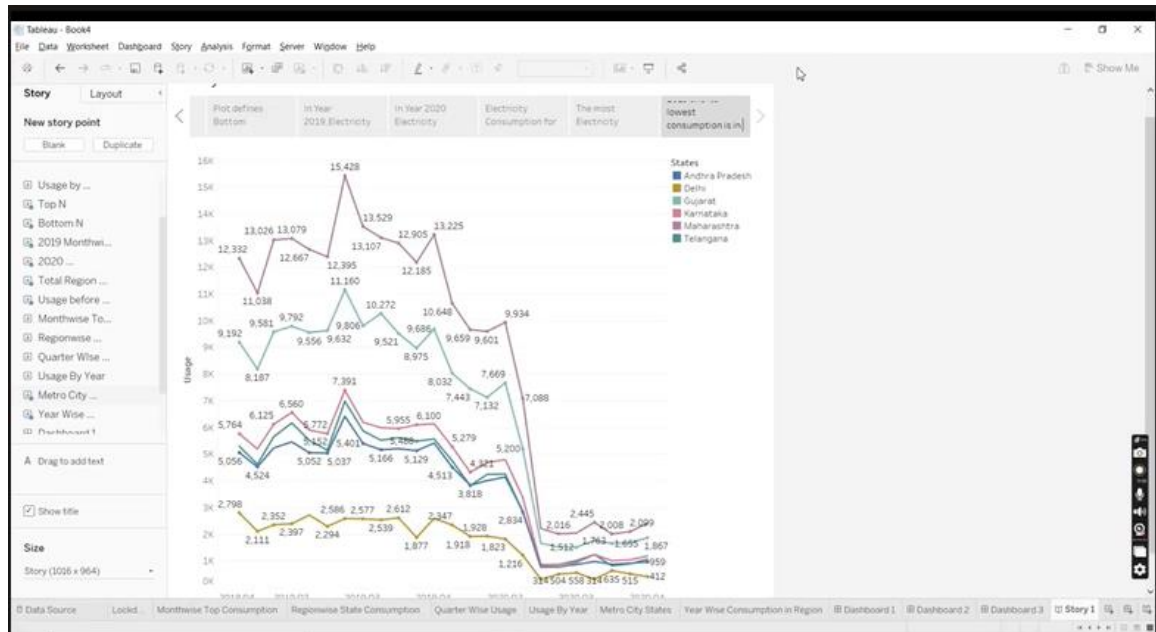
Quarter-wise Electricity Consumption.

Dashboard 4



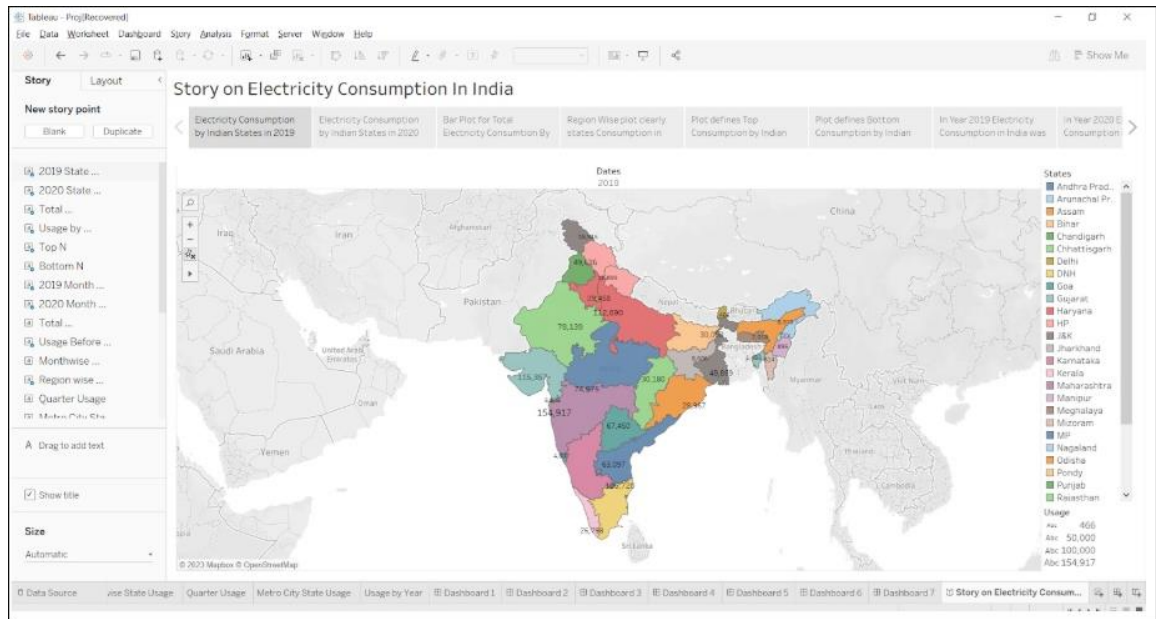
Region-wise Electricity Usage.

Dashboard 5



Total Electricity Consumption by States.

Dashboard 6



Top 5 and Bottom 5 Electricity Consuming States.

Dashboard 7

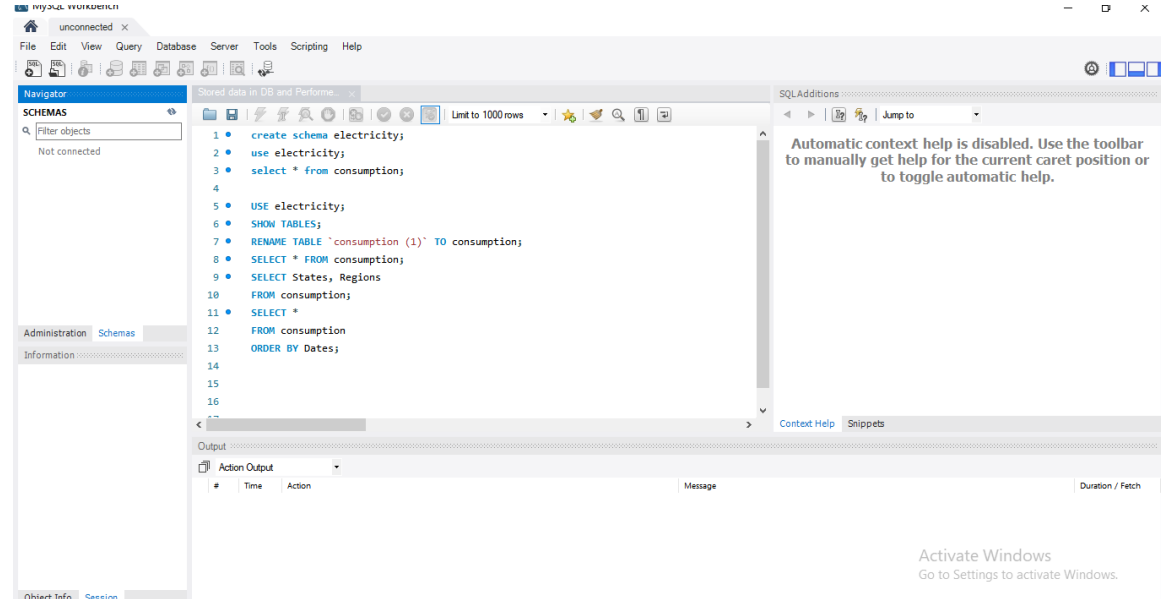


Tableau Story illustrating insights.

Dashboard 8



MySQL Database Queries.

8. ADVANTAGES & DISADVANTAGES

Advantages

- Interactive dashboards.
- Easy to understand.
- Quick comparison.
- Supports decision-making.

- Attractive visualization.
- Saves reporting time.
- Dynamic filtering.
- Better regional analysis.
- Real-time dashboard updates (when connected to live data).
- Easy dashboard sharing using Tableau Public.

Disadvantages

- Requires Tableau software.
 - Historical data only.
 - Depends on data quality.
 - Limited forecasting.
 - Requires database connectivity.
-

9. CONCLUSION

This project successfully demonstrates Business Intelligence implementation using MySQL and Tableau. The dashboards effectively analyze electricity consumption across Indian states and regions for the years 2019 and 2020. Interactive maps, charts, and stories enable users to identify consumption trends, compare yearly performance, and support informed decision-making. The project highlights how visualization tools can simplify complex datasets and improve reporting efficiency.

10. FUTURE SCOPE

The project can be further enhanced by incorporating:

- Real-time electricity data integration.
 - Machine Learning for demand forecasting.
 - AI-based anomaly detection.
 - Power outage prediction.
 - Web-based dashboard deployment.
 - Mobile application integration.
 - Weather-based electricity consumption analysis.
 - IoT sensor integration.
 - Predictive analytics.
 - Cloud database implementation.
 - Energy cost analysis.
 - Renewable energy consumption tracking.
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11. APPENDIX

MySQL Queries

```
CREATE DATABASE electricity;
```

```
USE electricity;
```

```
SHOW TABLES;
```

```
SELECT * FROM consumption;
```

```
SELECT States, Regions  
FROM consumption;
```

```
SELECT *  
FROM consumption  
ORDER BY Dates;
```

Dataset Fields

- State
- Region
- Date
- Month
- Quarter
- Year
- Usage

Tableau Components Used

- Maps
- Pie Chart
- Line Chart
- Bar Chart
- Dashboard
- Story
- Filters
- Parameters
- Legends
- Tool tips

Software Used

- MySQL Workbench
- Tableau Desktop
- Tableau Public
- Microsoft Excel

- Windows Operating System

GitHub & Project Demo Link :

<https://github.com/Anjum966/Electricity-Consumption>

<https://youtube.com/@junnusyed-s5s3n?si=OGfIOkWwgE8T90eI>